NVDA NVIDIA Corporation — 个股深度研究 (专业中文版) v20261016-0756

> 数据截至 2026-10-16 07:56 SGT｜价格 $217.23｜市值 5,245.5 bn USD｜财报货币 USD｜非投资建议。事实=CONFIRMED/DATA；推断=INFERENCE；假设=ASSUMPTION/HYPOTHESIS。

1. 投资论点 Investment Thesis (HYPOTHESIS)

NVDA 是 AI 资本开支的首要受益者，系统级(GPU+NVLink+网络+CUDA)护城河使其维持高毛利；核心争议在于云厂 Capex 持续性与 ASIC 替代速度。

- **多头驱动**: Rubin cycle on time, inference demand broadens to enterprise/sovereign, networking attach rises｜Rubin 按期、推理需求扩散、网络附加率提升
- **空头驱动**: Capex digestion in 2027, ASIC share gains, gross-margin pressure from HBM costs｜2027 年 Capex 消化、ASIC 份额提升、HBM 成本压毛利
- **证伪条件 Falsifiers**: Top-4 hyperscaler capex guides down; DC revenue q/q growth <5% two quarters; GM below mid-60s%｜四大云 Capex 下修；DC 收入环比连续两季<5%；毛利率跌破 60%中段

2. 产品、客户与收入模式

- 主题 Theme: AI compute GPU AI 算力
- 业务分部 Segments: Data Center compute 数据中心计算；Data Center networking 网络；Gaming 游戏；ProViz/Auto/Other 专业可视化/汽车
- 公司描述 (DATA, Yahoo/公司申报): NVIDIA Corporation operates as a data center scale AI infrastructure company in the United States, Taiwan, China, Hong Kong, Europe, and internationally. It operates through Compute & Networking, and Graphics segments. The Compute & Networking segment provides data center accelerated computing and networking platforms and artificial intelligence solutions and software, and automotive platforms and autonomous and electric vehicle solutions, including software. The Graphics segment offers GeForce GPUs for gaming and PCs; Quadro/NVIDIA RTX GPUs for enterprise workstation graphics. The company’ s products are used in gaming, professional visualization, data center, and automotive markets. It sells its products to original equipment manufacturers, original device manufacturers, system integrators and

distributors, independent software vendors, cloud service providers, add-in board manufacturer

- 主要客户: Hyperscalers (MSFT, META, GOOGL, AMZN, ORCL), neoclouds (CRWV), sovereigns, enterprises｜超大规模云厂、新云、主权 AI、企业

3. 行业、竞争、监管与供应链 (ANALYST FRAMEWORK)

- 供应商/上游: TSMC (N4/N3, CoWoS), SK hynix/Micron/Samsung (HBM), ODMs (Foxconn, Quanta, Wistron), ABF/PCB｜台积电(制程+CoWoS)、HBM 三巨头、ODM、ABF/PCB
- 竞争者 (含业外): AMD Instinct, custom ASICs (Broadcom/Marvell/Google TPU/Trainium), Intel, Chinese domestic GPUs｜AMD、定制 ASIC(TPU/Trainium 等)、英特尔、中国国产 GPU
- 瓶颈/监管: CoWoS capacity, HBM supply, data-center power, export controls (China)｜CoWoS 产能、HBM 供应、数据中心电力、出口管制
- 关键 KPI: DC revenue, GM%, networking revenue, hyperscaler capex, CoWoS allocation｜数据中心收入、毛利率、网络收入、云厂 Capex、CoWoS 分配
- 零部件现货/合约价: 本版未接入付费价格源(TrendForce/DRAMeXchange 等)——标注 **UNAVAILABLE**，在 KPI 监控中以定性方式跟踪。

4. 历史财务与现金流 (CONFIRMED, 季度, USD)

季度	收入	毛利	营业利润	净利润	FCF	营业利润率
2026-07-31	96.2 bn USD	72.1 bn USD	63.7 bn USD	59.7 bn USD	21.4 bn USD	66.2%
2026-04-30	81.6 bn USD	61.2 bn USD	53.5 bn USD	58.3 bn USD	48.6 bn USD	65.6%
2026-01-31	68.1 bn USD	51.1 bn USD	44.3 bn USD	43.0 bn USD	34.9 bn USD	65.0%
2025-10-31	57.0 bn USD	41.8 bn USD	36.0 bn USD	31.9 bn USD	22.1 bn USD	63.2%
2025-07-31	46.7 bn USD	33.9 bn USD	28.4 bn USD	26.4 bn USD	13.5 bn USD	60.8%

资产负债: 现金 62.5 bn USD；总负债 38.9 bn USD；净现金 23.6 bn USD。TTM FCF 127.0 bn USD（FCF 率 41.9%）。

5. 最新财报: 实际 vs 一致预期 vs 指引

季度	EPS 实际	EPS 一致预期	超预期%
2025-10-31	1.3	1.25647	0.0346
2026-01-31	1.62	1.53812	0.053200003
2026-04-30	1.87	1.77191	0.0554
2026-07-31	2.22	2.09113	0.0616

- 管理层指引 Guidance: 自动数据源不含指引——**UNAVAILABLE（待从公司 IR 新闻稿录入）**。
- 下次财报日 (Yahoo, TENTATIVE 直至公司确认): **2026-11-18**；本季 EPS 一致预期 2.47325（44 位分析师）；收入一致预期 109.0 bn USD。

6. 未来 2/4/6/8 季度驱动与价格合理性

期限	情景	期限价格现值	相对现价
2Q	Bear	$110.68	-49.1%
2Q	Base	$276.35	27.2%
2Q	Bull	$481.66	121.7%
4Q	Bear	$118.04	-45.7%
4Q	Base	$338.98	56.0%
4Q	Bull	$636.40	193.0%
6Q	Bear	$125.88	-42.1%
6Q	Base	$415.86	91.4%
6Q	Bull	$840.61	287.0%
8Q	Bear	$134.22	-38.2%
8Q	Base	$510.22	134.9%
8Q	Bull	$1,109.99	411.0%

现价隐含的 EV/NTM-FCF (Base 路径):	期限
2Q	18.4x
4Q	14.1x
6Q	10.8x
8Q	8.3x

解读: 如果 8 季度后 Base 情景所需倍数仍远高于 25x，说明价格依赖更远期的增长。

7. 三情景 Bear / Base / Bull (HYPOTHESIS)

情景	Y1 增速	长期 FCF 率	WACC	永续 g	概率	每股价值	相对现价
Bear	34.1%	36.0%	14.2%	2.5%	25.0%	$126.49	-41.8%
Base	68.2%	45.0%	13.2%	3.0%	50.0%	$551.38	153.8%
Bull	88.6%	51.7%	12.7%	3.5%	25.0%	$1,317.63	506.6%

概率加权价值 **$636.72**（相对现价 193.1%）。

8. 估值: DCF / Reverse DCF / ROIC-WACC / 同业 / 敏感度

- DCF Base: EV 13,290.6 bn USD；终值占比 55.9%。
- **Reverse DCF 反向 DCF**: 在 Base 利润率与 WACC 下，现价隐含 Y1 收入增速 ≈ **39.7%**；一致预期 FY+1 增速 68.2%。
- WACC (CAPM) = rf 5.2% + β 1.60 × ERP 5.0% = 13.2%。
- SOTP segments 分部: Data Center compute 数据中心计算; Data Center networking 网络; Gaming 游戏; ProViz/Auto/Other 专业可视化/汽车. **Status: framework only** — segment revenues are INPUT REQUIRED from the latest 10-K/10-Q segment note (not auto-fed, left blank to avoid fabrication)｜分部收入需按最新财报手动录入，未自动填充以避免编造。
- 同业比较 Comps (DATA): | Ticker | 市值 | EV/Sales | Fwd P/E | 收入增速 | 毛利率 | FCF 收益率 | |—|—|—|—|—|—|—| | NVDA | 5,245.5 bn USD | 17.2x | 13.6x | 105.9% | 74.7% | 2.4% | | AMD | 917.8 bn USD | 22.0x | 35.7x | 50.1% | 55.7% | 1.0% | | AVGO | 1,643.7 bn USD | 18.8x | 17.8x | 85.5% | 75.5% | 1.9% | | MRVL | 216.8 bn USD | 22.5x | 33.1x | 36.5% | 52.2% | 1.1% | | INTC | 504.1 bn USD | 9.1x | 45.3x | 25.4% | 38.9% | 1.0% |
- 敏感度: 见 Excel `Sensitivity` 表 (WACC × 永续增长率)。

9. 风险、反方论证与证伪

- 反方观点 Counter-thesis: Capex digestion in 2027, ASIC share gains, gross-margin pressure from HBM costs｜2027 年 Capex 消化、ASIC 份额提升、HBM 成本压毛利

- 证伪指标: Top-4 hyperscaler capex guides down; DC revenue q/q growth <5% two quarters; GM below mid-60s%｜四大云 Capex 下修；DC 收入环比连续两季<5%；毛利率跌破 60%中段
- 模型风险: 长期利润率与 WACC 为假设；终值占比高时估值对假设极敏感。

10. Source Ledger 来源账本

Item	Value	Source/time
Price 股价	217.2	Yahoo Finance quote, fetched 2026-10-16T07:56 SGT
Market cap 市值	5.245e+12	Yahoo info.marketCap 2026-10-16T07:56
Cash 现金	6.247e+10	Yahoo info.totalCash (latest filing) 2026-10-16T07:56
Debt 负债	3.886e+10	Yahoo info.totalDebt (latest filing) 2026-10-16T07:56
Revenue TTM 收入	3.03e+11	Yahoo info.totalRevenue (TTM, USD) 2026-10-16T07:56
FCF TTM 自由现金流	1.27e+11	Sum of last 4 quarterly FCF (q_cash); quarters ['2026-07-31', '2026-04-30', '2026-01-31', '2025-10-31'] (USD)
Beta	2.216	Yahoo info.beta (5Y monthly)
Risk-free 10Y 无风险利率	0.0519	FRED DGS10 2026-10-14
FX USD per USD	1	n/a (USD)
Consensus rev growth FY0	0.9064	Yahoo revenue_estimate 0y
Consensus rev growth FY+1	0.6815	Yahoo revenue_estimate +1y

11. Assumption Register 假设登记表

Assumption	Value	Rationale
ERP 股权风险溢价	0.05	ASSUMPTION: 5.0% house mature-market

Assumption	Value	Rationale
		ERP
Beta clamp	[0.8,1.6] → 1.60	ASSUMPTION: limit extreme betas
Tax rate (ROIC)	0.18	ASSUMPTION: blended cash tax 18%
Year-1 growth (Base)	0.6815	Consensus FY+1 revenue growth (Yahoo revenue_estimate '+1y'); capped at 100%
Growth fade	linear Y1→terminal by Y10	ASSUMPTION
Starting FCF margin	0.4192	DATA: FCF TTM / Revenue TTM, floored -50%
Long-run FCF margin (Base)	0.45	ASSUMPTION (analyst framework, reached by Y5; see profile)
Bear/Bull Y1 growth multiplier	0.5x / 1.3x	ASSUMPTION
Bear/Bull margin multiplier	0.8x / 1.15x	ASSUMPTION
WACC adj Bear/Bull	+1.0pp / -0.5pp	ASSUMPTION
Terminal g Bear/Base/Bull	2.5% / 3.0% / 3.5%	ASSUMPTION
Scenario probabilities	25% / 50% / 25%	ASSUMPTION
Horizon exit EV/NTM-FCF Bear/Base/Bull	15x / 25x / 35x	ASSUMPTION
Per-share conversion	value × fx ÷ market cap × price	METHOD: avoids ADR/share-class mismatch

12. Model Provenance 模型溯源

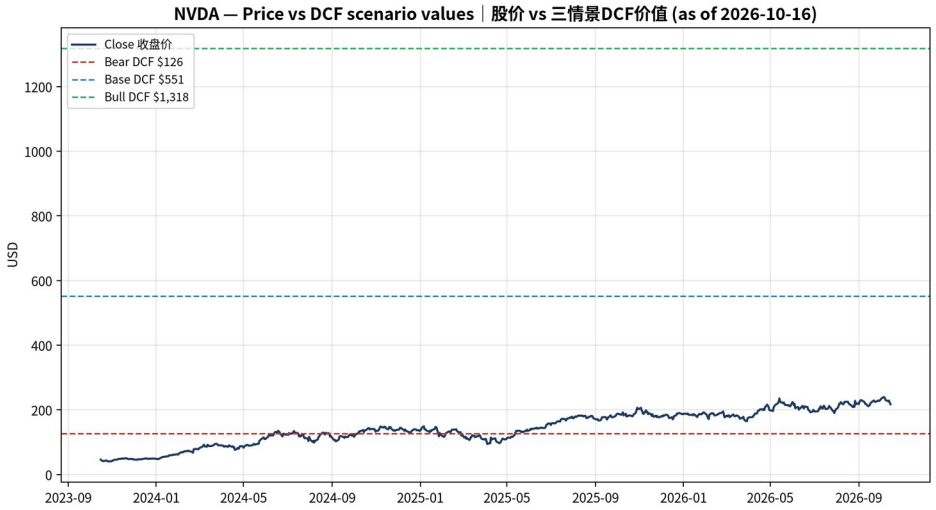
- Model file: `NVDA_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions)
- Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx)
- Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT

- Version: v20261016-0756. Prior versions are never overwritten (see company history page).

13. 版本与事件时间轴

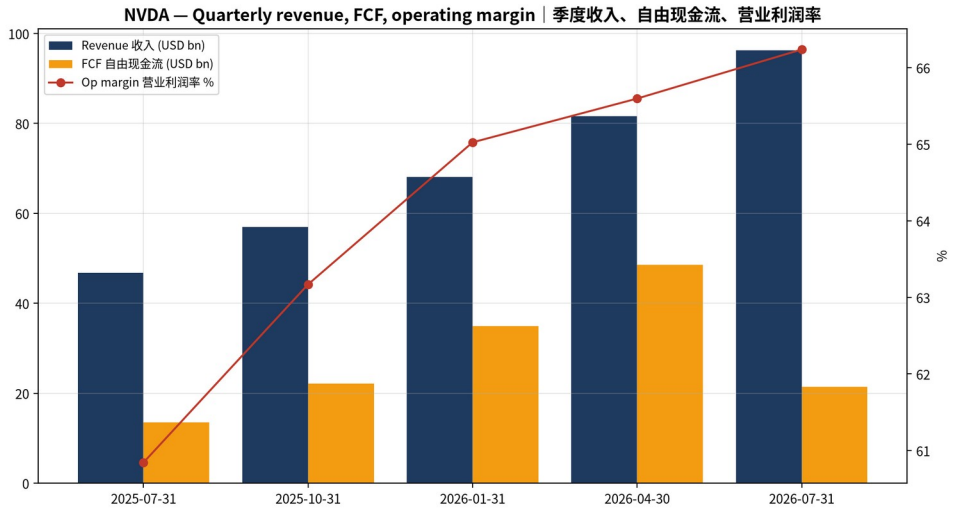
- v20261016-0756: 首版生产系统报告 (27 家同步)。下一事件: 财报 2026-11-18 (TENTATIVE)。

图片报告 Image Report



1_price_vs_value

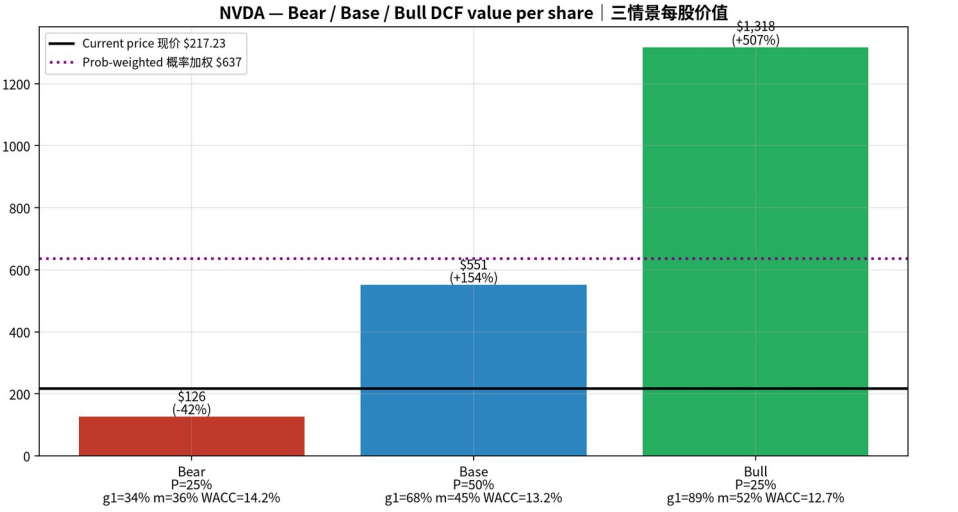
股价vs 三情景价值Price vs scenario values



Claim class 声明类别: CONFIRMED (reported) | Source 来源: Company filings via Yahoo Finance quarterly statements
Thinking 思考: Is FCF growing with revenue? Divergence = capex/working-capital intensity → check cash-flow statement. | Not investment advice 非投资建议

2_financials

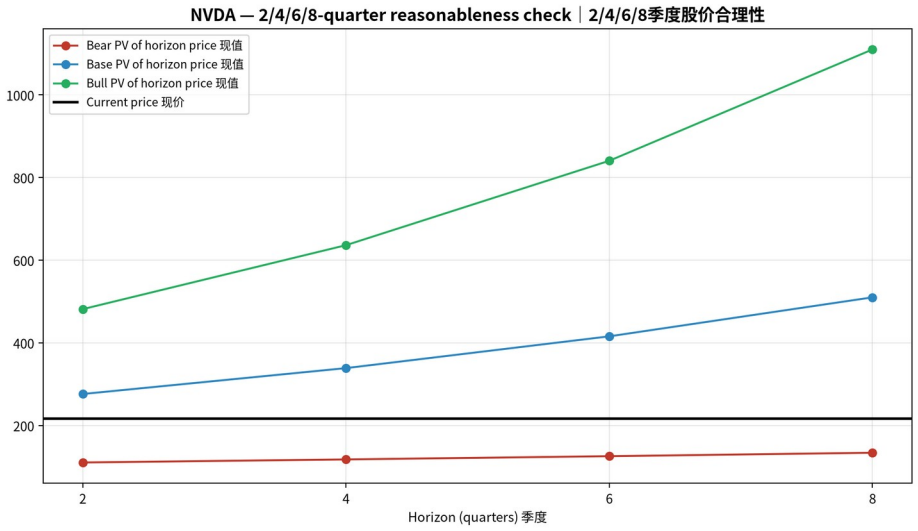
季度财务 Quarterly financials



Claim class 声明类别: HYPOTHESIS (model output) | Source 来源: Model (live formulas) — inputs in Sources/Assumptions sheets
Thinking 思考: Which assumption moves value most? Check Sensitivity sheet; scenario width = uncertainty. | Not investment advice 非投资建议

3_scenarios

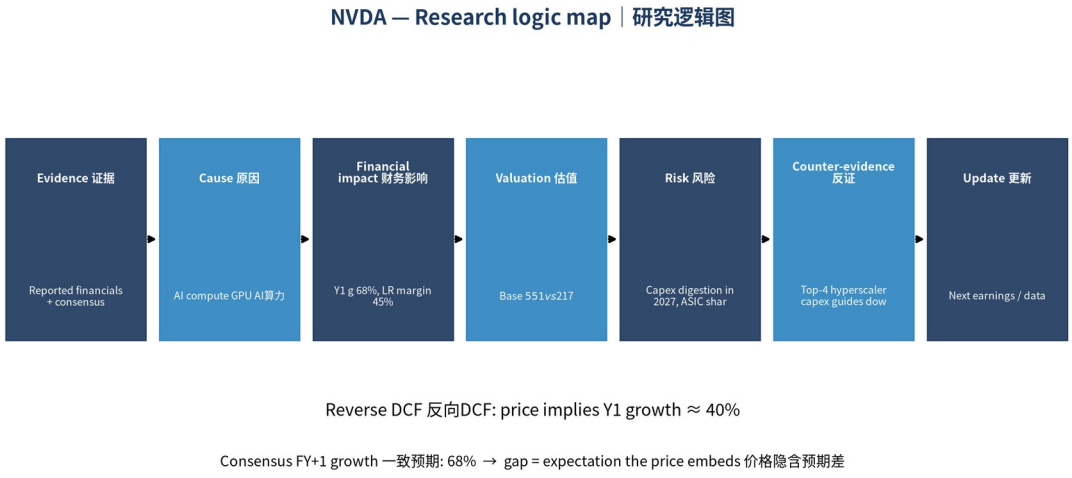
三情景 Scenarios



Claim class 声明类别: HYPOTHESIS | Source 来源: Model Horizons sheet; exit EV/NTM-FCF 15x/25x/35x (ASSUMPTION)
Thinking 思考: If even Bull PV < price at 8Q, price requires assumptions beyond Bull within 2 years. | Not investment advice 非投资建议

4_horizons

2/4/6/8 季度检验Horizon check



Reverse DCF 反向DCF: price implies Y1 growth ≈ 40%

Consensus FY+1 growth 一致预期: 68% → gap = expectation the price embeds 价格隐含预期差

Claim class 声明类别: INFERENCE / HYPOTHESIS | Source 来源: Profile (analyst framework) + model
Thinking 思考: Read left→right: each box must have evidence; the falsifier box tells you what would prove the thesis wrong. | Not investment advice 非投资建议

5_logic_map

研究逻辑图Logic map

NVDA NVIDIA Corporation — Equity Deep Dive (Professional English) v20261016-0756

Data as of 2026-10-16 07:56 SGT | Price $217.23 | Mkt cap 5,245.5 bn USD | Reporting currency USD | Not investment advice.

1. Investment thesis (HYPOTHESIS)

NVDA is the primary beneficiary of AI capex; its system-level moat (GPU+NVLink+networking+CUDA) sustains high margins. The core debate is hyperscaler capex durability and the pace of ASIC substitution.

- **Bull drivers**: Rubin cycle on time, inference demand broadens to enterprise/sovereign, networking attach rises｜Rubin 按期、推理需求扩散、网络附加率提升
- **Bear drivers**: Capex digestion in 2027, ASIC share gains, gross-margin pressure from HBM costs｜2027 年 Capex 消化、ASIC 份额提升、HBM 成本压毛利
- **Falsifiers**: Top-4 hyperscaler capex guides down; DC revenue q/q growth <5% two quarters; GM below mid-60s%｜四大云 Capex 下修；DC 收入环比连续两季<5%；毛利率跌破 60%中段

2. Products, customers, revenue model

- Theme: AI compute GPU AI 算力 | Segments: Data Center compute 数据中心计算; Data Center networking 网络; Gaming 游戏; ProViz/Auto/Other 专业可视化/汽车
- Business description (DATA): NVIDIA Corporation operates as a data center scale AI infrastructure company in the United States, Taiwan, China, Hong Kong, Europe, and internationally. It operates through Compute & Networking, and Graphics segments. The Compute & Networking segment provides data center accelerated computing and networking platforms and artificial intelligence solutions and software, and automotive platforms and autonomous and electric vehicle solutions, including software. The Graphics segment offers GeForce GPUs for gaming and PCs; Quadro/NVIDIA RTX GPUs for enterprise workstation graphics. The company’s products are used in gaming, professional visualization, data center, and automotive markets. It sells its products to original equipment manufacturers, original device manufacturers, system integrators and distributors, independent software vendors, cloud service providers, add-in board manufacturer
- Customers: Hyperscalers (MSFT, META, GOOGL, AMZN, ORCL), neoclouds (CRWV), sovereigns, enterprises｜超大规模云厂、新云、主权 AI、企业

3. Industry, competition, regulation, supply chain (ANALYST FRAMEWORK)

- Suppliers: TSMC (N4/N3, CoWoS), SK hynix/Micron/Samsung (HBM), ODMs (Foxconn, Quanta, Wistron), ABF/PCB｜台积电(制程+CoWoS)、HBM 三巨头、ODM、ABF/PCB
- Competitors: AMD Instinct, custom ASICs (Broadcom/Marvell/Google TPU/Trainium), Intel, Chinese domestic GPUs｜AMD、定制 ASIC(TPU/Trainium 等)、英特尔、中国国产 GPU
- Bottlenecks/regulation: CoWoS capacity, HBM supply, data-center power, export controls (China)｜CoWoS 产能、HBM 供应、数据中心电力、出口管制
- KPIs: DC revenue, GM%, networking revenue, hyperscaler capex, CoWoS allocation｜数据中心收入、毛利率、网络收入、云厂 Capex、CoWoS 分配
- Component spot/contract prices: paid feeds not connected — **UNAVAILABLE**, tracked qualitatively.

4. Historical financials & cash flow (CONFIRMED, quarterly, USD)

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2026-07-31	96.2 bn USD	72.1 bn USD	63.7 bn USD	59.7 bn USD	21.4 bn USD	66.2%
2026-04-30	81.6 bn USD	61.2 bn USD	53.5 bn USD	58.3 bn USD	48.6 bn USD	65.6%
2026-01-31	68.1 bn USD	51.1 bn USD	44.3 bn USD	43.0 bn USD	34.9 bn USD	65.0%
2025-10-31	57.0 bn USD	41.8 bn USD	36.0 bn USD	31.9 bn USD	22.1 bn USD	63.2%
2025-07-31	46.7 bn USD	33.9 bn USD	28.4 bn USD	26.4 bn USD	13.5 bn USD	60.8%

Cash 62.5 bn USD; debt 38.9 bn USD; TTM FCF 127.0 bn USD (FCF margin 41.9%).

5. Latest earnings: actual vs consensus vs guidance

Quarter	EPS actual	EPS consensus	Surprise
2025-10-31	1.3	1.25647	0.0346
2026-01-31	1.62	1.53812	0.053200003
2026-04-30	1.87	1.77191	0.0554

Quarter	EPS actual	EPS consensus	Surprise
2026-07-31	2.22	2.09113	0.0616

- Management guidance: not in automated feed — **UNAVAILABLE (to be entered from IR release)**.
- Next earnings (Yahoo; TENTATIVE until company-confirmed): **2026-11-18**; current-quarter EPS consensus 2.47325 (44 analysts); revenue consensus 109.0 bn USD.

6. 2/4/6/8-quarter drivers & price reasonableness

Horizon	Scenario	PV of horizon price	vs current
2Q	Bear	$110.68	-49.1%
2Q	Base	$276.35	27.2%
2Q	Bull	$481.66	121.7%
4Q	Bear	$118.04	-45.7%
4Q	Base	$338.98	56.0%
4Q	Bull	$636.40	193.0%
6Q	Bear	$125.88	-42.1%
6Q	Base	$415.86	91.4%
6Q	Bull	$840.61	287.0%
8Q	Bear	$134.22	-38.2%
8Q	Base	$510.22	134.9%
8Q	Bull	$1,109.99	411.0%

Current-price implied EV/NTM-FCF on Base path:	Horizon
2Q	18.4x
4Q	14.1x
6Q	10.8x
8Q	8.3x

7. Bear / Base / Bull (HYPOTHESIS)

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bear	34.1%	36.0%	14.2%	2.5%	25.0%	$126.49	-41.8%
Base	68.2%	45.0%	13.2%	3.0%	50.0%	$551.38	153.8%
Bull	88.6%	51.7%	12.7%	3.5%	25.0%	$1,317.6	506.6%

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/share	vs price
						3	

Probability-weighted value **$636.72** (193.1% vs price).

8. Valuation: DCF / Reverse DCF / ROIC-WACC / comps / sensitivity

- Base DCF EV 13,290.6 bn USD; terminal share 55.9%.
- **Reverse DCF**: at Base margin/WACC the price implies Y1 revenue growth ≈ **39.7%** vs consensus FY+1 68.2%.
- WACC (CAPM) = rf 5.2% + β 1.60 × ERP 5.0% = 13.2%.
- SOTP segments: Data Center compute 数据中心计算; Data Center networking 网络; Gaming 游戏; ProViz/Auto/Other 专业可视化/汽车 — framework only; segment revenue inputs pending from filings.
- Comps (DATA): see table in Chinese section / Excel `Comps`.
- Sensitivity: Excel `Sensitivity` (WACC × terminal g).

9. Risks, counterarguments, falsifiers

Capex digestion in 2027, ASIC share gains, gross-margin pressure from HBM costs｜2027 年 Capex 消化、ASIC 份额提升、HBM 成本压毛利 | Falsifiers: Top-4 hyperscaler capex guides down; DC revenue q/q growth <5% two quarters; GM below mid-60s%｜四大云 Capex 下修；DC 收入环比连续两季<5%；毛利率跌破 60% 中段

10–12. Source Ledger / Assumption Register / Model Provenance

See tables above (shared, bilingual). - Model file: `NVDA_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions) - Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx) - Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT - Version: v20261016-0756. Prior versions are never overwritten (see company history page).